[19] Sensing - Battery

DESIGN NOTE:
Battery voltage sensing divider for 2S LiPo battery.

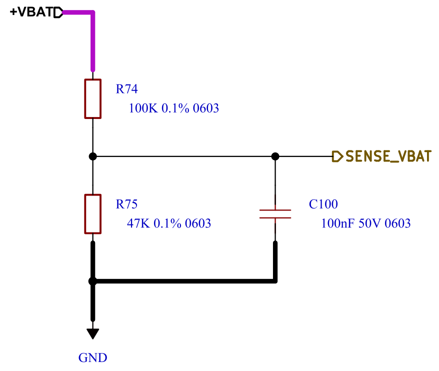
R1 = 100kΩ (Top)
R2 = 47kΩ (Bottom)

Divider Ratio:
VADC = VBAT × (47 / (100 + 47))
VADC = VBAT × 0.3197

2S LiPo:
Max battery voltage (8.4V): 2.69 V
Nominal voltage (7.4V): 2.37 V
Low battery voltage (6.4V): 2.05 V
Cutoff voltage (6.0V): 1.92 V

Maximum divider current:
I = 8.4V / 147kΩ = 57 µA

Notes:
- Designed for 3.3V ADC input.
- Add 100nF capacitor at ADC input for noise filtering.
- Use 1% resistors or better.
- Use low temperature coefficient resistors (≤100 ppm/°C) for improved measurement accuracy.



Battery Voltage Sensing

Comments:	Company: Willas (威倫電子股份有限公司 Willas Electronic Corp.)		Variant: DRAFT	Git Hash: 538a591
	Board Name: **Motor Control Board**		Project Name: **Master Hand**	
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